Lancelot Juarez

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Objective

Mechanical Engineering student with hands-on experience in PCB design, prototyping, testing, and control systems through rocket and UAV projects at UC San Diego. Skilled in SolidWorks, Python, and C++, and interested in applying these skills to energy storage and sustainability technologies.

Experience

SEDS@UCSD | *Avionics & Controls Lead, VESPER Rocket* June 2025 – Present

- Designed a custom 8 channel DAQ PCB for rocket instrumentation.
- Developed hardware and implemented PI controls for throttling a rocket 650 lbf LOX/LCH4 film-cooled engine.
- Troubleshot hardware and integration issues during subsystem testing.
- Prepared technical documentation for design reviews and team coordination.
- Developed and Validated a REDS (Rapid Emergency Depressurization System).

RPL@UCSD | *Daedalus Lead Avionics & Analysis Engineer* Sep 2024 – June 2025

- Evaluated G-class rocket stability, structural design, and predicted flight performance using SolidWorks and OpenRocket.
- Designed and used a custom avionics PCB for onboard data acquisition.
- Preformed post launch analysis achieving an apogee of .

Projects

Autonomous UAV Wildfire Detection System Aug 2024 – June 2025

- Built a fully 3D-printed fixed-wing UAV platform.
- Integrated embedded hardware for onboard wildfire detection.
- Implemented a YOLO-based vision model for real-time fire detection.
- Programmed autonomous flight paths using Raspberry Pi and MAVLink.
- Troubleshot hardware, software, and communication system issues.

Education

University of California, San Diego 2024 – Expected 2027
Bachelor of Science in Mechanical Engineering
Specialization in Controls and Robotics

Skills

- **Technical Skills:** SolidWorks, Onshape, Git, MATLAB, OpenRocket, PCB Design, Simulink, Embedded Systems
- **Interpersonal Skills:** Prototyping, Testing, Validation, Leadership in team based Environments
- **Programming:** Python, C++, Arduino, Java, HTML, Shell Scripting

Last updated: April 23, 2026